

Estimating the constant value

$$x(k)=x(k+1)=0.12345$$

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Using the measurement $z(k)=x(k)+w(k)$ where w is a white noise $w \sim N(0,0.01)$, we'll implement Kalman's filter algorithm for discrete time.

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Create the noise and data

create a random sequence of numbers. function randn draws numbers from standard normal distribution.

```
r=0.01;
noise=randn(1,150);
% create the random white noise sequence by multiplying the set of random
% numbers by the %standard deviation (tipiki apoklisi) of w.
w=sqrt(r)*noise;
% create the measurements vector
zmeasure=0.12345+w;
% create the real-time constant vector
xconstant1(1:150)=0.12345;
```

The kalman filter algorithm (first 150 estimations)

```
xprior(1)=0;
Pprior(1)=1;
K(1)=Pprior(1)/(Pprior(1)+r);
```

```

        xpost(1)=xprior(1)+K(1)*(zmeasure(1)-xprior(1));
        Ppost(1)=(1-K(1))*Pprior(1);
    for i=2:150
        xprior(i)=xpost(i-1);
        Pprior(i)=Ppost(i-1);
        K(i)=Pprior(i)/(Pprior(i)+r);
        xpost(i)=xprior(i)+K(i)*(zmeasure(i)-xprior(i));
        Ppost(i)=(1-K(i))*Pprior(i);
    end
    k=1:150;

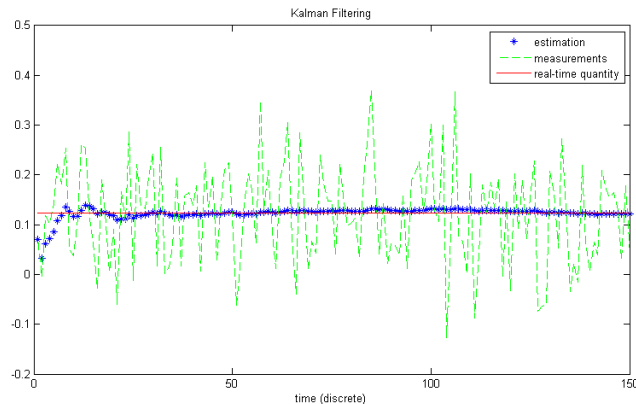
```

Plotting the results

```

fig = figure;
set(fig, 'Position', [100 100 900 500]);
% The above two commands help determine the size and position of the figure
plot(k,xpost(k),'*',k,zmeasure(k),'g--',k,xconstant1(k),'r')
legend('estimation','measurements','real-time quantity')
xlabel 'time (discrete)'
title 'Kalman Filtering'

```



References

- [1] Technical staff, the analytic sciences corporation, edited by Arthur Gelb (1974). Applied Optimal Estimation. The MIT Press.
- [2] Feng Lin (2007). Robust Control Design, An optimal control approach. John Wiley and Sons.
- [3] Alok Sinha (2007). Linear Systems, Optimal and Robust Control. CRC Press.

[4] Lazaros Moysis 2012, The Kalman Filter (In Greek), Assignment for the M.Sc. course "Optimal Control of systems". <http://users.auth.gr/lazarosm/KalmanFilter-MoysisLazaros.pdf>